Exploit Walkthrough:
Verizon Ellipsis Jetpack MHS800L

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Feburary 2026

Overview

We have analyzed the binary and found a potential command injection exploit, CWE-78. Please reference "Iot Device Security Analysis: Verizon Ellipsis Jetpack mHS800L" for a detailed breakdown.

Books:

The Internet of Risky Things: Trusting the Devices That Surround Us

The Hardware Hacking Handbook: Breaking Embedded Security with Hardware Attacks

Tools:

Ghidra, OFRAK, Luadeck

IoT Devices:

Verizon Ellipsis Jetpack

Model Number: MHS800L

Authorized Remote Code Execution (RCE)

Authorized RCE is an exploit that requires a secret to run. In this case, we know the Wi-Fi password, which is the Admin password.

In the wild, we can grab the password in a few different ways.

- WPA2 Cracking:
  - Listen to all Wi-Fi traffic
  - Send a De-authentication packet to a legitimate user
  - When the victim automatically reconnects the password hash is sent
  - Grab this hash then crack it with a tool like Hashcat (need GPUs)
- Man In The Middle (MITM):
  - Victim is connected to Wi-Fi
  - Malicious actor broadcasts a fake hotspot with the same SSID
  - Victim's computer connects to the malicious actor's Wi-Fi
  - Fake admin portal login is prompted on the victim machine
  - If the password is entered, the password is routed to the hacker
- Sniffing:
  - HTTP not HTTPS, so no encryption.
  - If the victim logs into the WebUI a malicious actor can intercept the traffic
  - Grab Victim's real cookie
  - Use cookie and send modified HTTP Request
- Physical Access:
  - The password is displayed on the LED screen
  - In a real pen test, an insider would need to give this information away
- Gray Box Pen Testing:
  - A reverse engineer is contracted by a company to analyze an IoT device
  - Gray Box means the company provides information such as the firmware/binary or logins
  - In this case, the company provides the Wi-Fi password

Since this is a simulation, we have the WiFi password at our disposal. Grab the password from the LED display

The Walkthrough

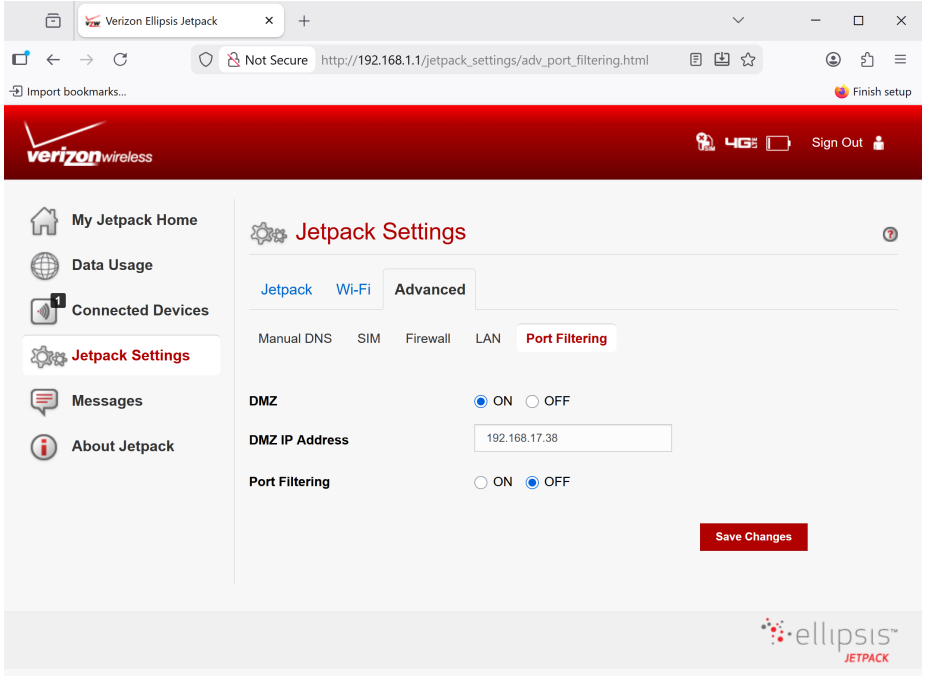

Figure 1: Visualizing Jetpack's WebUI, and where the DMZ settings are located

Using Figure 1 as a visual guide, on a computer connect to the Jetpack. Then open a browser and go to (192.168.1.1). I will use Firefox, but most browsers are similar. Navigate to Jetpack Settings - Advanced - Port Filtering.

Here we can see the input box for the DMZ request we saw earlier. To recap, the user input will travel from the WebUI, directly into the BusyBox shell without being checked. This means we can try a command injection attack to use telnetd to open a port that we will log into.

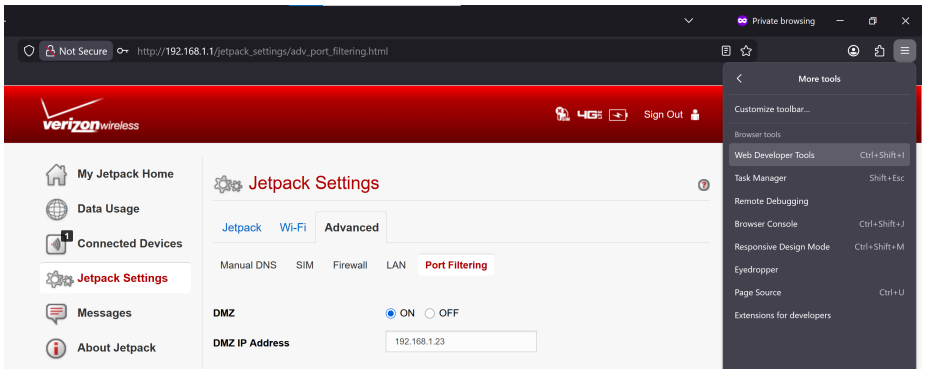

Figure 2: Illustrating how to get into Web Developer Tools

Using Figure 2 as a walkthrough, let's first grab the HTTP request using the developer tools in the browser. We can find this by clicking the three buttons in the top right corner, then navigating to **More Tools**, where we will find **Web Developer Tools**.

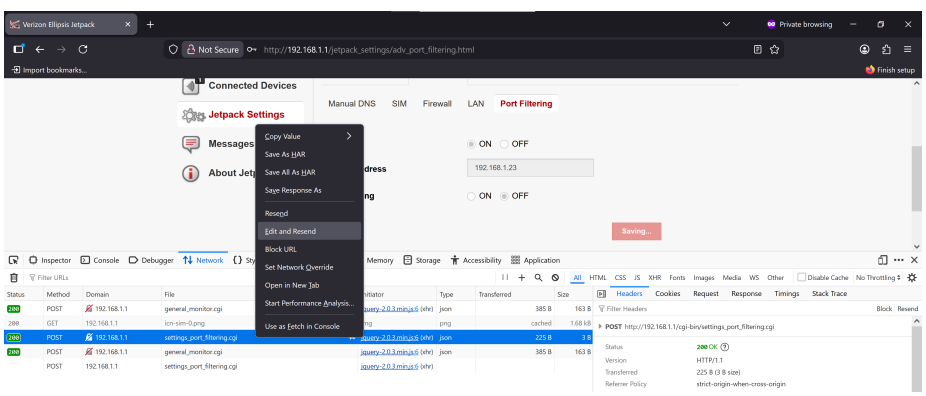

Figure 3: Showcasing the Network tab, and Edit and Resend Function

Using Figure 3 as a guide, with the developer window open, navigate to the **Network** tab. This shows our HTTP Requests. Now, in the Jetpack portal, turn DMZ on and set a bogus IP address. After clicking save, the HTTP POST should be sent. We should see a **POST with "settings_port_filtering.cgi"**. Click Edit and Resend (Copy Fetch on Google), then look for the body of the HTTP POST.

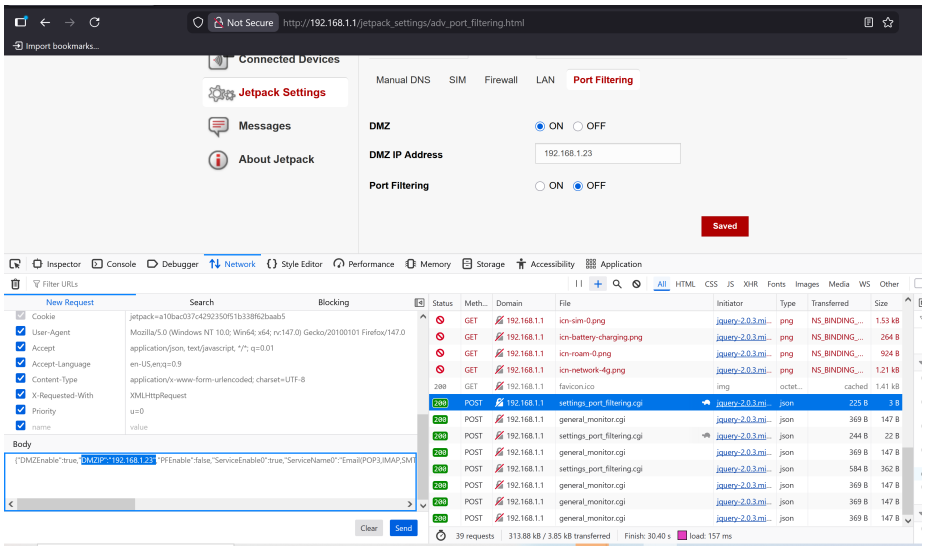

Figure 4: Visualizing where the unmodified HTTP POST payload is

We care about the field where **DMZIP** is set. This is where we can pass input directly to BusyBox.

```
{"DMZEnable":true,"DMZIP":"192.168.1.23","PFEnable":.......}
```

```
Let's Inject:
    $(telnetd -l /bin/sh -p6767)
```

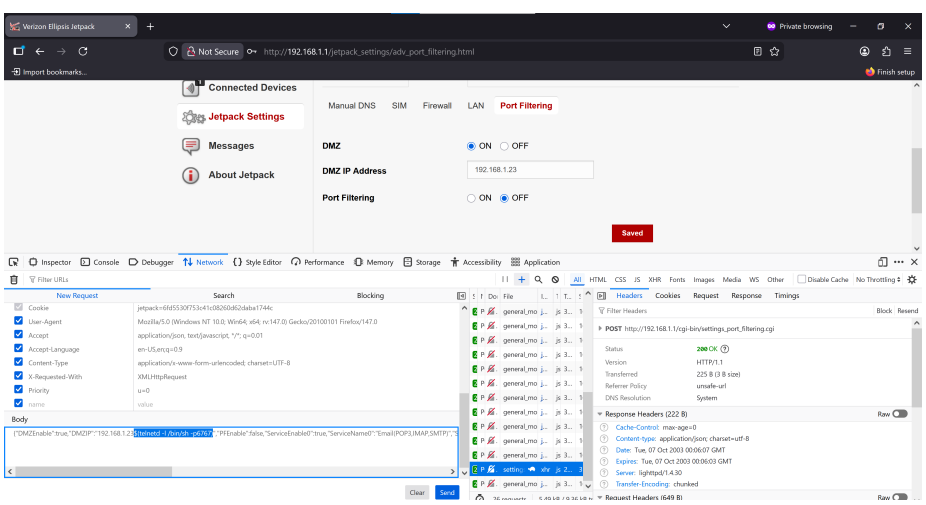

Figure 5: Illustrating modification to HTTP POST payload to inject command

```
{...,"DMZIP":"192.168.1.23$(telnetd -l /bin/sh -p6767)","PFEnable": ...)
```

Modify, then click send. Now let's try to connect to the port.

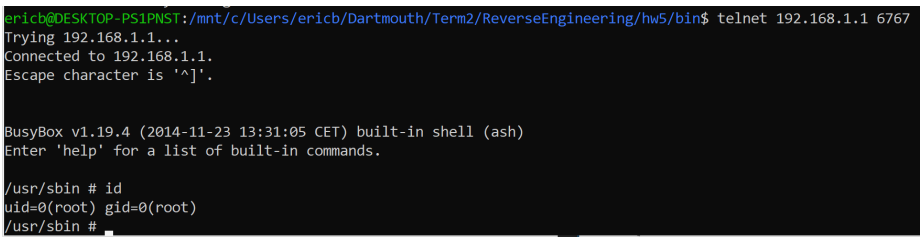

Figure 6: Reverse Shell into IoT device as root

As shown in Figure 6, the command injection worked and we now have a shell.

Next Steps:

Establish Foothold: If the jetpack resets, the shell is gone. Establishing persistence is the highest priority.

Internal Recon: In the cyber kill chain, the next step would be discovering what devices are connected to the jetpack because we are behind the firewall now. Since this is a lab, there will be nothing.

Lateral Movement: In a real network, other devices would be connected to the hotspot. They would be discovered after doing reconnaissance. To attack the other devices, we can setup a proxy or port forwarding on the IoT device to go around the firewall.

Data Exfiltration The router won't have much useful information, the devices connected will. However, the DNS configuration can be modified to point to a bogus website, such as a fake bank login controlled by a hacker.